

a Experimental Paradigm



"the big blue circle sticker"
target = dashed frame · 6 objects

Factorial Design (size-colour)

REFERENTIAL CONTEXT · within-subj.

Size

size alone

Both

size + colour

Colour

colour alone

SIZE DISCRIMINABILITY · between-subj.

High

strong contrast

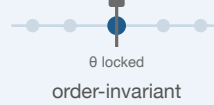
Low

weak contrast

b Incremental RSA Model

COMPOSITIONAL SEMANTICS · LISTENER L_0

Context-Fixed



Context-Updating



does word order matter? "big blue N" vs "blue big N"

PRAGMATIC SPEAKER S_1 · PRODUCTION

Global

big blue N

scored as one unit
whole utterance

Incremental

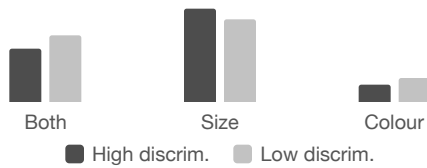


$$\text{utility } U = \alpha \cdot (\text{informativeness} - \text{cost})$$

c Empirical Results

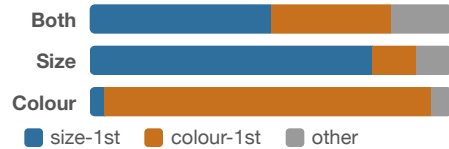
Exp 1 · Slider Rating

mean size-first preference



Exp 2 · Free Production

proportion by utterance category



d Model Comparison & Key Findings

2 × 2 Comparison (PSIS-LOO)

Key Findings

	Context-Fixed	Context-Updating
Global	G × Fixed baseline	G × Updating
Incremental	I × Fixed ★ best	I × Updating near tie

- Q1 **Context-updating** can generate baseline preference.
- Q2 **Incremental production** explains behaviour once order and salience are included.